

Elementary Problems Extremal Combinatorics

Version 1.0

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Basic Work

1. There are 50 cards. The numbers 1 to 100 are written randomly on different sides of the cards so that each side of a card contains a number. We can flip the cards to choose the sides that face upwards. Let S be the sum of numbers written on the cards which face upwards. Find the maximum k such that we can always ensure $S \geq k$ regardless of how the numbers are written on the cards.
2. There are 80 lockers. Each locker can be opened by a distinct kind of key. These keys are kept by 100 people. Different people may have the same kind of keys. Whenever any 10 of the people have left, the keys kept by the remaining 90 people are sufficient to open all the lockers. What is the minimum value of the total number of keys kept by all people?
3. We put 111 coins in the cells of an $n \times n$ board such that each cell may contain any number of coins (possibly 0). If the numbers of coins in two adjacent cells (sharing a common edge) differ by exactly 1, find the maximum n .
4. (CSMO 2004 Problem 4) Given a positive integer n larger than 2004, we put the numbers $1, 2, 3, \dots, n^2$ in the cells of an $n \times n$ chessboard (formed by n rows and n columns of cells) such that each cell contains exactly one number. If the number in a cell is larger than the numbers in at least 2004 cells in the same row, and larger than the numbers in at least 2004 cells in the same column, then we say that this cell is a 'good cell'. Find the greatest possible value of the number of 'good cells' in the chessboard.
5. There is a $3 \times 3 \times 3$ wooden block. Each time we can cut the wooden block along a plane into two halves. Also, we are allowed to place the wooden blocks formed over the others in each cut. At least how many cuts are needed to form 27 unit cubes at the end?
6. (Bosnia and Herzegovina TST 2011 Problem 4) Find the maximum value of number a such that for any arrangement of numbers $1, 2, \dots, 10$ on a circle, we can find three consecutive numbers such that their sum is bigger than or equal to a .
7. There are $n \geq 3$ cards, each with a black side and a white side. Initially, all cards have their black sides facing upwards. Each step we can flip over exactly 3 cards. At least how many steps are needed so that all cards have their white sides facing upwards?
8. (IMO shortlist 1979) Let $n \geq 2$ be an integer. Find the maximal cardinality of a set M of pairs (j, k) of integers, $1 \leq j < k \leq n$, with the following property: If $(j, k) \in M$, then $(k, m) \notin M$ for any m .

9. Initially, the integers from 1 to n are placed in the positions 1 to n in order respectively. We permute the integers in any way we want. The movement of an integer is the difference of the positions it is placed before and after the permutation. Find the greatest possible sum of the movements of all integers.
10. (Baltic Way 1990) Numbers $1, 2, \dots, n$ are written around a circle in some order. What is the smallest possible sum of the absolute differences of adjacent numbers?
11. Let $A_1, A_2, \dots, A_m$ be subsets of $\{1, 2, \dots, 8\}$ such that each subset has exactly 3 elements and $|A_i \cap A_j| \leq 1$ for any $i \neq j$. Find the greatest possible value of m .
12. Let $n \geq 3$. Let $A_1, A_2, \dots, A_n$ be pairwise distinct nonempty sets, where each of them is a subset of the union of the other sets. Find the minimum value of $|A_1| + |A_2| + \dots + |A_n|$.
13. (Pan African 2009 Problem 4) Consider n children in a playground, where $n \geq 2$. Every child has a coloured hat, and every pair of children is joined by a coloured ribbon. For every child, the colour of each ribbon held is different, and also different from the colour of that child's hat. What is the minimum number of colours that needs to be used?
14. Let n and k be positive integers. We write down the integers from 1 to 2^n on the two sides of k pieces of paper so that each integer appears exactly once on each paper. Suppose we can choose one side of each paper so that each integer from 1 to 2^n appears on the side that we choose of at least one piece of paper. Find the smallest k for which this is always possible regardless of how the integers are written down.
15. (Argentina TST 2011 Problem 6) Each 1×1 square of an $n \times n$ grid is coloured using red or blue, in such a way that among all the 2×2 subgrids, there are all the possible colourings of a 2×2 grid using red or blue (colourings that can be obtained by rotation or symmetry are said to be different, so that there are 16 possibilities).
 - (a) Find the minimum value of n .
 - (b) For that value, find the least possible number of red squares.

Pigeonhole Principle

16. At least how many numbers should be chosen from $1, 2, \dots, 100$ so that there must be 3 chosen numbers which are consecutive?
17. Let $n \geq 4$ be a positive integer. Find the minimum value of k such that among any k vertices of a regular n -gon, there are always 4 consecutive vertices of the regular $4n$ -gon.
18. Let n be a positive integer. At most how many subsets of $\{1, 2, 3, \dots, n\}$ can we choose so that no two chosen subsets are disjoint?

19. The integers 1 to 1000 are written on the board. We erase k integers randomly. Find the maximum k such that there are always 4 integers remaining with sum 1000 no matter how we erase the integers.
20. Find the smallest positive integer k with the following property: among any k integers, we can always find two of them whose sum or difference is a multiple of 100.
21. (IMO shortlist 1978) Let S be the set of all the odd positive integers that are not multiples of 5 and that are less than $30m$, m being an arbitrary positive integer. What is the smallest integer k such that in any subset of k integers from S there must be two different integers, one of which divides the other?
22. Let $n \geq 2$ be a positive integer. Each vertex of a regular $2n$ -gon is painted using one of the k colours. Find the smallest possible value of k such that no matter how we paint the vertices with all colours being used, there must be 3 vertices with pairwise distinct colours forming a right-angled triangle.
23. Let $n \geq 2$ be a positive integer. At most how many cells of an $n \times n$ table can be chosen so that no two of the chosen cells have centres lying in the same diagonal line? (A diagonal line is any line making an angle of 45° with the horizontal.)
24. At most how many distinct integers can be chosen from $1, 2, \dots, 11$ so that no two of them have difference 4 or 7?
25. Let n be a positive integer. Find the minimum k such that every k -element subset of $\{1, 2, 3, \dots, 2n\}$ contains 4 elements whose sum is $4n + 1$.
26. (Turkey 2014) $3m$ balls numbered $1, 1, 1, 2, 2, 2, 3, 3, 3, \dots, m, m, m$ are distributed into 8 boxes so that any two boxes contain identical balls. Find the minimal possible value of m .
27. (CGMO 2015 Problem 3) In a 12×12 grid, colour each unit square with either black or white, such that there is at least one black unit square in any 3×4 and 4×3 rectangle bounded by the grid lines. Determine, with proof, the minimum number of black unit squares.
28. (Poland MO Finals 2015 Problem 3) Find the biggest natural number m that has the following property: among any five 500-element subsets of $\{1, 2, \dots, 1000\}$ there exist two sets, whose intersection contains at least m numbers.
29. (Erdős-Ko-Rado theorem) Let n and m be positive integers such that $n \geq 2m$. Let A be a family of subsets of $\{1, 2, \dots, n\}$ such that each subset in A contains m elements and any pair of subsets in A has a nonempty intersection. Find the maximum value of $|A|$.

Induction

30. Let n be a positive integer. There are 3 rods, one with n disks of increasing size from top to bottom, and the other two being empty. Each step we can move the disk at the top of one rod to another rod, with the restriction that the size of this disk must be smaller than all the disks in the new rod. At least how many steps are needed to transfer all the n disks to another rod?
31. Let $n \geq 4$ be an integer. Given a polygon with n sides, we draw some diagonals of the polygon one by one. Each time the diagonal drawn can only intersect at most one other diagonal which has been drawn in the interior (diagonals meeting at the vertices of the polygon are not counted). Find the maximum number of diagonals which may be drawn. (Note that one can choose how to draw the diagonals.)
32. (Macedonia MO 2013 Problem 2) 2^n coins are given to a couple of kids. Interchange of the coins occurs when some of the kids has at least half of all the coins. Then from the coins of one of those kids to all the other kids are given that much coins as the kid already had. In case when all the coins are at one kid there is no possibility for interchange. What is the greatest possible number of consecutive interchanges?
33. (EGMO 2012 Problem 2) Let n be a positive integer. Find the greatest possible integer m , in terms of n , with the following property: a table with m rows and n columns can be filled with real numbers in such a manner that for any two different rows $[a_1, a_2, \dots, a_n]$ and $[b_1, b_2, \dots, b_n]$ the following holds:
- $$\max \{|a_1 - b_1|, |a_2 - b_2|, \dots, |a_n - b_n|\} = 1.$$
34. Let $m, n \geq 2$ be integers. At most how many rooks can be put on an $m \times n$ chessboard so that each rook is threatened by exactly 2 other rooks? The rooks threaten each other if they are put on two cells in the same row or the same column and there is no other rook between them.
35. (USAMO 2000 Problem 4) Find the smallest positive integer n such that if n squares of a 1000×1000 chessboard are coloured, then there will exist three coloured squares whose centres form a right triangle with sides parallel to the edges of the board.
36. (IMO shortlist 2003 C3) Let $n \geq 5$ be a given integer. Determine the greatest integer k for which there exists a polygon with n vertices (convex or not, with non-selfintersecting boundary) having k internal right angles.
37. (IMO shortlist 1966) An alphabet consists of n letters. What is the maximal length of a word if we know that any two consecutive letters a, b of the word are different and that the word cannot be reduced to a word of the kind $abab$ with $a \neq b$ by removing letters?

Record

v1.0: problems 1–37